

Research Problem

Goal

– The purpose of Hydra Vision was to find lost objects using the TurtleBot4, a small autonomous robot based off the Roomba. To perform this task, we needed an object detection model that would be efficient and accurate for the Turtlebot, given the strengths and limitations of the hardware.



Contribution

What We Did

- Our team developed python based scripts that would log our comparison metrics while the simulation was running. The script would run 300 image frames against each model.
- Our experiments were conducted in the TurtleBot Simulator, in Gazebo Harmonic, a robot simulator.

Proposed Method

- We would be using three object detection models, YOLOv8, Faster R-CNN, and SSD (single shot detection) and running each in a simulated environment with the TurtleBot4.

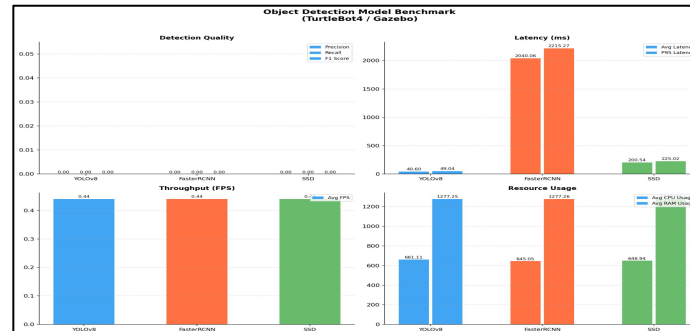
Comparison Metrics:

- mAP (Mean Average Precision)
- Average FPS (Frames per Second)
- CPU/GPU usage
- Precision/Recall

Experimental Results

Our Experiment

- We were not attempting to find the “best model”, rather to see each model’s strengths and weaknesses.
- Each simulated environment was the same for each model.
- Our simulations were run through Ubuntu on a VM.



Conclusion

Our Findings

- Difficulties with the simulation software prevented us from obtaining results for the accuracy of our models. However, given the role our device is supposed to fulfill, this was of secondary concern compared to computational efficiency.
- YOLOv8 was clearly a more responsive model, but it came at the cost of more computational draw.
- Faster R-CNN is a very slow model, but cost the least. However, its savings was not dramatic compared to the other models.

Reference

Libraries/Tools Used:

- YOLOv8
- TurtleBot4 Simulation
- Gazebo Harmonic
- Faster R-CNN
- SSD
- Python3